NEET (UG) Official Syllabus & Statutory Chapter Weightage

ISSUED BY NATIONAL MEDICAL COMMISSION (NMC) & NATIONAL TESTING AGENCY (NTA)

Examination Standard: Total Marks: 720 | Total Questions: 200 (180 to Attempt) | Duration: 200 Minutes
Marking Scheme: +4 for Correct Response, -1 for Incorrect, 0 for Unattempted.
Statutory Section Allocation: Physics: 45 Qs (180 M) • Chemistry: 45 Qs (180 M) • Botany: 45 Qs (180 M) • Zoology: 45 Qs (180 M).
Official Curriculum Notice: Strictly verified against latest NMC Under Graduate Medical Education Board guidelines including rationalized and added topics.

Subject: Physics (18 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NMC/NTA)
Units, Dimensions & Errors Physical World and Measurement	Class 11	Active (Physical World historical intro deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	SI units, fundamental/derived units, dimensional analysis, screw gauge, vernier calipers, error propagation.
Motion in a Straight Line & Plane Kinematics	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**High Yield**	1D uniformly accelerated motion, graphs (x-t, v-t), relative velocity, projectile motion trajectory, uniform circular motion.
Newton's Laws & Friction Laws of Motion	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Inertia, momentum, Newton's laws, impulse, conservation of linear momentum, static/kinetic friction, banking of roads.
Work, Energy, Power & Collisions Work, Energy and Power	Class 11	Active	**2 Qs** 8 Marks	**Guaranteed Core**	Work-energy theorem, spring potential energy, conservative forces, power, 1D elastic and inelastic collisions.
System of Particles & Rotational Dynamics Rotational Motion	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Centre of mass, torque, angular momentum conservation, moment of inertia (ring, disc, cylinder, sphere), parallel/perpendicular axes theorems.
Gravitation & Satellites Gravitation	Class 11	Active	**2 Qs** 8 Marks	**Guaranteed Core**	Kepler's laws, universal gravitation, variation of g with height/depth, gravitational PE, escape speed, orbital speed.
Solids, Fluids & Thermal Properties Properties of Bulk Matter	Class 11	Active	**3 - 4 Qs** 12 - 16 Marks	**High Yield**	Hooke's law, Young's modulus, Pascal's law, hydraulic lift, viscosity, Stokes' law, terminal velocity, Bernoulli's theorem, surface tension, capillary rise, calorimetry, Newton's law of cooling.
Thermodynamics & Kinetic Theory Thermodynamics & KTG	Class 11	Active (Carnot Engine deleted by NMC/NTA)	**3 Qs** 12 Marks	**Guaranteed Core**	1st law of thermodynamics, isothermal, adiabatic, isobaric, isochoric processes, work done in PV cycles, KTG pressure derivation, RMS velocity, degrees of freedom, molar heat capacities.
SHM & Wave Motion Oscillations and Waves	Class 11	Active (Doppler effect deleted by NMC/NTA)	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	SHM kinematics, restoring force, simple pendulum, energy in SHM, transverse/longitudinal waves, standing waves in organ pipes (open/closed), resonance, beats.
Electric Charges, Fields & Potential Electrostatics	Class 12	Active (Van de Graaff deleted)	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Coulomb's law, electric field lines, electric dipole in uniform field, Gauss's law applications, electrostatic potential, equipotential surfaces, potential energy of charge configurations.
Capacitors & Dielectrics Electrostatics	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Capacitance, parallel plate capacitor with/without dielectric, series and parallel combination, energy stored in capacitor, charge redistribution.
Electric Current, Resistance & Circuits Current Electricity	Class 12	Active (Potentiometer & Colour codes deleted)	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Drift velocity, mobility, Ohm's law, resistivity, temperature dependence of resistance, series/parallel combinations, EMF, internal resistance, cells in series/parallel, Kirchhoff's laws, Wheatstone bridge, Metre bridge.
Moving Charges & Magnetism Magnetic Effects of Current	Class 12	Active (Cyclotron & Earth magnetic elements deleted)	**3 Qs** 12 Marks	**Guaranteed Core**	Biot-Savart law, circular coil field, Ampere's circuital law, solenoid field, force on charge in B-field, force on parallel current conductors, torque on coil, moving coil galvanometer conversion to ammeter/voltmeter, magnetic properties (dia-, para-, ferro-).

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NMC/NTA)
EMI, Alternating Currents & Transformers Electromagnetic Induction & AC	Class 12	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Faraday's laws, Lenz's law, motional EMF, self/mutual inductance, AC generator, peak/RMS values, series LCR circuit, resonance, power factor, transformer.
Electromagnetic Spectrum & Displacement Current Electromagnetic Waves	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Displacement current concept, transverse nature of EM waves, electromagnetic spectrum order (radio to gamma rays) and specific medical/technological applications.
Ray Optics, Optical Instruments & Wave Optics Optics	Class 12	Active (Polarization deleted)	**4 - 5 Qs** 16 - 20 Marks	**Guaranteed Core**	Refraction at spherical surfaces, lens maker's formula, thin lenses in contact, total internal reflection, prism deviation, compound microscope, astronomical telescope, Huygens' principle, interference (YDSE fringe width), single slit diffraction central maximum.
Dual Nature, Atoms & Nuclei Modern Physics	Class 12	Active (Radioactivity decay law deleted by NMC/NTA)	**4 - 5 Qs** 16 - 20 Marks	**Guaranteed Core**	Photoelectric effect, Einstein's equation, de Broglie matter waves, Bohr's model of H-atom, spectral lines (Lyman, Balmer, Paschen), nuclear composition, mass defect, binding energy per nucleon curve, nuclear fission and fusion.
Semiconductor Electronics Electronic Devices	Class 12	Active (Transistors, Logic Gates & Zener Diode deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Energy band gap in solids, intrinsic and extrinsic semiconductors (p-type, n-type), p-n junction diode forward and reverse bias characteristics, diode as half-wave and full-wave rectifier.

Subject: Chemistry (7 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NMC/NTA)
Mole Concept, Atomic Structure & Bonding Physical Chemistry	Class 11	Active	**4 - 6 Qs** 16 - 24 Marks	**Guaranteed Core**	Stoichiometry, molarity/molality/mole fraction, empirical formula, Bohr model, quantum numbers, electronic configurations, VSEPR shapes, hybridization (sp to sp3d2), dipole moments, MOT bond orders.
Thermodynamics & Equilibrium Physical Chemistry	Class 11	Active	**4 - 5 Qs** 16 - 20 Marks	**Guaranteed Core**	1st and 2nd laws, enthalpy of formation/combustion, Hess's law, Gibbs free energy spontaneity, chemical equilibrium (Kp, Kc, Le Chatelier), ionic equilibrium (pH, buffer, salt hydrolysis, Ksp).
Solutions, Electrochemistry & Kinetics Physical Chemistry	Class 12	Active (Solid State & Surface Chem deleted)	**5 - 7 Qs** 20 - 28 Marks	**Guaranteed Core**	Raoult's law, colligative properties, van't Hoff factor, Nernst equation, cell potential, Kohlrausch's law, Faraday's electrolysis, reaction rate laws, 1st order integrated equations, Arrhenius equation.
Periodic Classification & p-Block Elements Inorganic Chemistry	Class 11	Active (s-Block, Hydrogen, Metallurgy deleted)	**3 - 4 Qs** 12 - 16 Marks	**High Yield**	Periodic trends (IE, EA, radius, electronegativity), group 13 to 18 electronic configurations and general chemical trends, anomalous behaviour of second period elements, inert pair effect.
d- and f-Block Elements & Coordination Compounds Inorganic Chemistry	Class 12	Active	**5 - 6 Qs** 20 - 24 Marks	**Guaranteed Core**	Transition elements properties, oxidation states, KMnO4 and K2Cr2O7 preparations/reactions, lanthanoid contraction, Werner's theory, IUPAC nomenclature of complexes, VBT and CFT splitting, magnetic moments, isomerism.
General Organic Chemistry (GOC) & Hydrocarbons Organic Chemistry	Class 11	Active (Environmental Chemistry deleted)	**4 - 5 Qs** 16 - 20 Marks	**Guaranteed Core**	IUPAC nomenclature, inductive, electromeric, resonance, hyperconjugation effects, carbocation/carbanion stability, electrophilic aromatic substitution, Markovnikov addition, ozonolysis, alkynes acidity.
Functional Organic Compounds & Biomolecules Organic Chemistry	Class 12	Active (Polymers & Everyday Life deleted)	**7 - 9 Qs** 28 - 36 Marks	**Guaranteed Core**	SN1/SN2 mechanisms, Lucas test, Reimer-Tiemann, Kolbe, Aldol condensation, Cannizzaro, Clemmensen/Wolff-Kishner, acidic strength of carboxylic acids, Hinsberg test, Gabriel phthalimide, carbohydrates (glucose, fructose), proteins (peptide bond, denaturation), nucleic acids (DNA, RNA).

Subject: Biology (Botany) (7 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NMC/NTA)
The Living World, Biological Classification & Plant Kingdom Diversity in Living World	Class 11	Active	**6 - 8 Qs** 24 - 32 Marks	**Guaranteed Core**	Binomial nomenclature, taxonomical aids (rationalized), 5-kingdom classification (Monera, Protista, Fungi), viruses/viroids, algae, bryophytes, pteridophytes, gymnosperms (alternation of generations).
Morphology & Anatomy of Flowering Plants Structural Organisation	Class 11	Active (Added Malvaceae, Cruciferae, Leguminosae, Compositae, Gramineae by NMC)	**5 - 7 Qs** 20 - 28 Marks	**Guaranteed Core**	Roots, stems, leaves modifications, inflorescence, flower anatomy, fruit types, placentation, floral formula and floral diagrams of families: Solanaceae, Malvaceae, Cruciferae (Brassicaceae), Leguminosae (Fabaceae), Compositae (Asteraceae), Gramineae (Poaceae); Anatomy: meristematic/permanent tissues, vascular bundles, secondary growth in dicots.
Cell: Unit of Life & Cell Cycle Cell Structure & Function	Class 11	Active	**6 - 8 Qs** 24 - 32 Marks	**Guaranteed Core**	Prokaryotic vs eukaryotic cell, cell wall, endomembrane system, mitochondria, chloroplasts, ribosomes, cytoskeleton, nucleus, mitosis, meiosis stages (prophase I leptotene to diakinesis), crossing over.
Photosynthesis, Respiration & Plant Growth Plant Physiology	Class 11	Active (Transport in Plants & Mineral Nutrition deleted)	**6 - 8 Qs** 24 - 32 Marks	**Guaranteed Core**	Light reaction, cyclic/non-cyclic photophosphorylation, Calvin C3 cycle, Hatch-Slack C4 pathway, photorespiration, glycolysis, TCA cycle, electron transport chain (chemiosmosis), plant growth regulators (auxin, gibberellin, cytokinin, ethylene, ABA).
Sexual Reproduction in Flowering Plants Reproduction	Class 12	Active (Reproduction in Organisms deleted)	**5 - 6 Qs** 20 - 24 Marks	**Guaranteed Core**	Microsporogenesis, megasporogenesis, pollen grain structure, embryo sac development (7-celled, 8-nucleate), pollination types, outbreeding devices, pollen-pistil interaction, double fertilization, endosperm and embryo development, apomixis.
Mendelian Genetics & Molecular Basis of Inheritance Genetics & Evolution	Class 12	Active (Highest-yielding biology cluster)	**10 - 14 Qs** 40 - 56 Marks	**Paramount Core**	Mendel's laws, incomplete dominance, co-dominance, multiple alleles, chromosomal theory, linkage and recombination, sex determination, pedigree analysis, DNA structure, packaging in nucleosomes, Meselson-Stahl experiment, replication, transcription, genetic code, translation, lac operon, human genome project, DNA fingerprinting.
Organisms, Populations, Ecosystem & Biodiversity Ecology & Environment	Class 12	Active (Environmental Issues deleted)	**6 - 8 Qs** 24 - 32 Marks	**Guaranteed Core**	Population attributes (birth rate, death rate, age pyramids), population growth models (exponential, logistic), population interactions (mutualism, competition, predation, parasitism), ecological pyramids, energy flow, productivity, biodiversity hotspots, in-situ and ex-situ conservation.

Subject: Biology (Zoology) (9 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NMC/NTA)
Animal Kingdom (Non-Chordata & Chordata) Diversity in Living World	Class 11	Active	**4 - 5 Qs** 16 - 20 Marks	**Guaranteed Core**	Levels of organisation, symmetry, coelom types, segmentation, notochord; Characteristic features and examples of phyla: Porifera, Coelenterata, Ctenophora, Platyhelminthes, Aschelminthes, Annelida, Arthropoda, Mollusca, Echinodermata, Hemichordata, Chordata (Cyclostomata, Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves, Mammalia).
Animal Tissues, Cockroach & Frog Anatomy Structural Organisation	Class 11	Active (Frog Rana tigrina explicitly added by NMC)	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Epithelial, connective, muscular and nervous tissues; Morphology, anatomy and digestive, circulatory, respiratory, nervous, and reproductive systems of Cockroach (Periplaneta americana) and Frog (Rana tigrina).
Biomolecules Cell Structure & Function	Class 11	Active	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Chemical constituents of living cells: carbohydrates, lipids, amino acids, proteins (primary to quaternary structure), nucleic acids, nature of enzyme action, factors affecting enzyme activity (temperature, pH, substrate concentration, competitive inhibition).
Breathing, Circulation, Excretion & Locomotion Human Physiology	Class 11	Active (Digestion and Absorption deleted)	**7 - 9 Qs** 28 - 36 Marks	**Guaranteed Core**	Respiratory volumes (TV, IRV, ERV, RV, VC), oxygen-haemoglobin dissociation curve, cardiac cycle, ECG waves (P, QRS, T), double circulation, nephron structure, urine formation, counter-current mechanism, sliding filament theory of muscle contraction, human skeletal system joints.

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NMC/NTA)
Neural Control & Chemical Coordination **Human Physiology**	Class 11	Active (Sensory organs eye/ear details rationalized)	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Generation and conduction of nerve impulse, synapse transmission, human endocrine system (hypothalamus, pituitary, thyroid, parathyroid, adrenal, pancreas, gonads), mechanisms of hormone action (peptide vs steroid hormones).
Human Reproduction & Reproductive Health **Reproduction**	Class 12	Active	**5 - 6 Qs** 20 - 24 Marks	**Guaranteed Core**	Male and female reproductive systems, spermatogenesis and oogenesis hormonal control, menstrual cycle phases, fertilization, cleavage, blastocyst implantation, placenta functions, parturition, lactation; Contraception methods, MTP, STIs, assisted reproductive technology (IVF, ZIFT, GIFT, ICSI).
Evolution **Genetics & Evolution**	Class 12	Active	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Origin of life, Miller-Urey experiment, evidence of evolution (homologous vs analogous organs, embryological), natural selection (industrial melanism), adaptive radiation (Darwin's finches), Hardy-Weinberg equilibrium, human evolution stages.
Human Health & Disease, Microbes in Human Welfare **Biology in Human Welfare**	Class 12	Active (Strategies in Food Production deleted)	**4 - 5 Qs** 16 - 20 Marks	**Guaranteed Core**	Common human diseases (malaria life cycle, typhoid, amoebiasis, pneumonia), immune system (innate, acquired, humoral, cell-mediated), active vs passive immunity, vaccination, AIDS (HIV replication cycle), cancer mechanisms, drug abuse, microbes in sewage treatment, biogas, biocontrol agents, biofertilizers.
Biotechnology Principles & Applications **Biotechnology**	Class 12	Active	**5 - 7 Qs** 20 - 28 Marks	**Guaranteed Core**	Restriction enzymes, cloning vectors (pBR322 selectable markers amp^R, tet^R), gel electrophoresis, PCR steps (denaturation, annealing, extension), bioreactors, genetically engineered insulin, Bt cotton mechanism, RNA interference (RNAi), gene therapy (ADA deficiency).